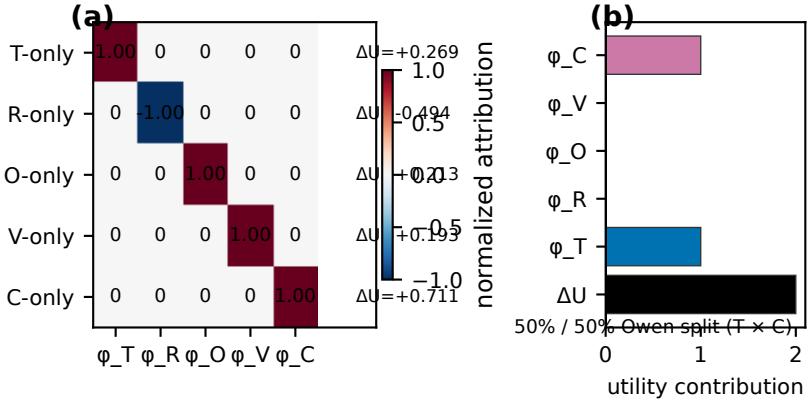
(a)
T-only
R-only
O-only
V-only
C-only
1.00
ΔU=+0.269
ΔU=-0.494
ΔU=+0.213
ΔU=+0.193
ΔU=+0.711
φ_T φ_R φ_O φ_V φ_C
1.0
0.5
−0.5
−1.0
(b)
φ_C
φ_V
φ_O
φ_R
φ_T
ΔU
normalized attribution
50% / 50% Owen split (T × C),
0
1
2
utility contribution
(c) checks
score → pattern 0 A/B
pattern/value → 0 A/B
masked probability
independent endpoint identity
all-A injected-z identity
all-B injected-z identity
non-candidate modification
live Q leak 0
live V leak 0
live Q+V leak 0
model quotient bridge
synthetic quotient bridge
partition coverage max
partition overlap max
overall PASS